

Experiment-5

Objective – Design a Data Flow Diagram (DFD) of Railway Reservation System.

A Data Flow Diagram (DFD) is a graphical tool used to represent how data moves through a system. It shows data inputs, outputs, data stores, and the processes that transform the data. DFDs provide a high-level overview of system functionality and are widely used in structured analysis due to their simplicity and clarity for both technical and non-technical users

- Input data
- Processing steps
- Data storage
- Output data

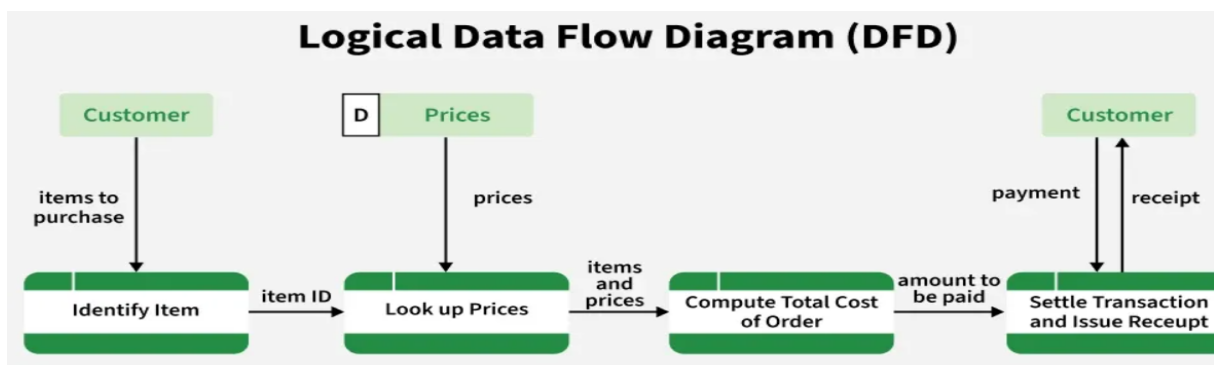
Basic Components of DFD

1. **Process (Circle / Bubble)**
Represents processing of data
Example: "Process Order"
2. **Data Flow (Arrow)**
Shows movement of data
Example: "Customer Details"
3. **Data Store (Open Rectangle)**
 - Stores data
 - Example: "Database"
4. **External Entity (Rectangle)**
 - Source or destination of data
 - Example: "Customer"

Types of DFD

1. **Level 0 (Context Diagram)**
 - Shows entire system as one process
2. **Level 1 DFD**
 - Breaks system into sub-processes
3. **Level 2 DFD**
 - Further detailed breakdown

Example: DFD for Railway Reservation System.



Explanation

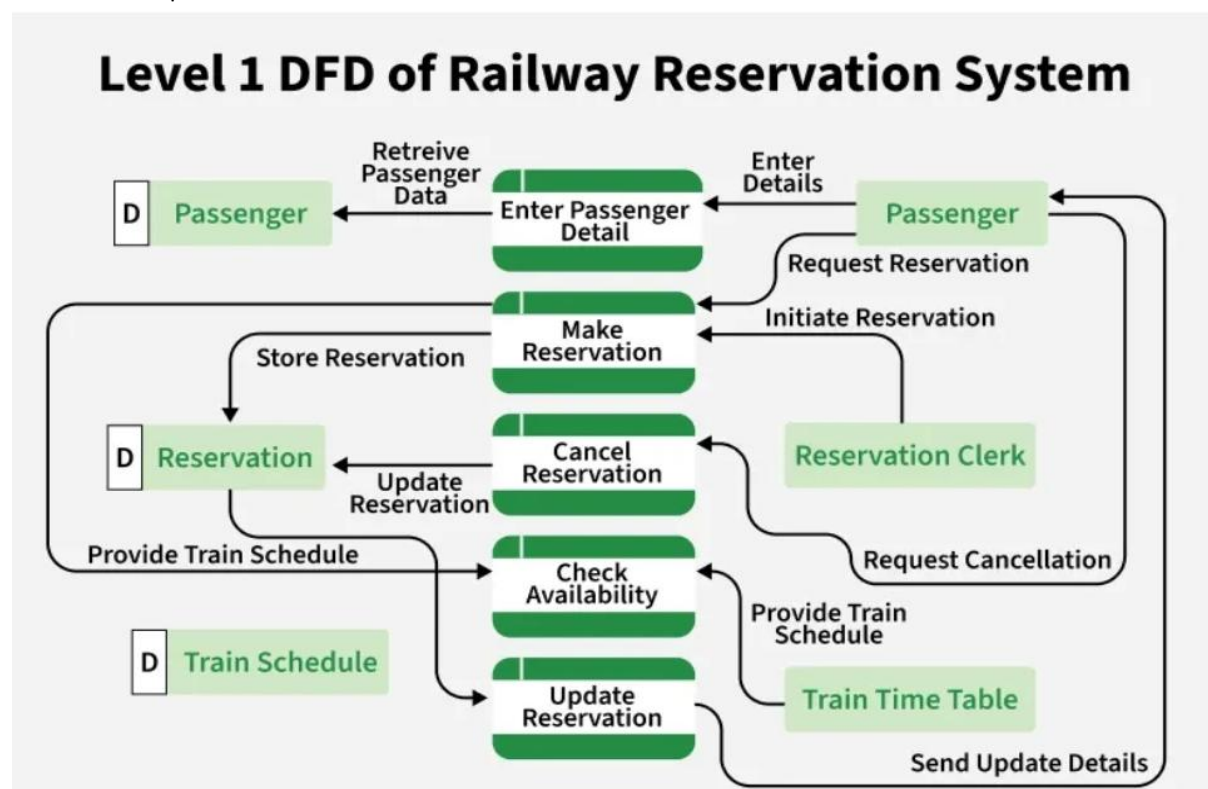
External Entity → Customer

Processes → Identify Item, Look up Prices, Compute Total Cost, Settle Transaction

Data Store → Prices (D)

Data Flow → Arrows showing movement of data

Level 1 DFD |



A Level 2 DFD is a detailed decomposition of Level 1 processes.

It breaks each major process into smaller sub-processes for deeper understanding.

Advantages of DFD

- Easy to understand
- Helps in system design
- Shows data movement clearly

- Useful for communication between developers and users

Disadvantages

- Does not show control flow
- Can become complex for large systems
- Not suitable for real-time systems